

SMU x LUSHPROTEIN

Mid-term Report

Prepared by: Group 1

Aditya Vijay

Ang Wei Lin

Emily Pham Vinh Tan

Saelin Lee

Shwe Tin Aung

Zheng Feng Toh

Contents

1. Data Overview and Coverage	3
1.1 All source data sets – Complete inventory	3
1.2 Canonical output tables	3
1.3 Temporal Coverage.....	4
1.4 Store and currency coverage	4
2. Summary Statistics and Distributions.....	5
3. Data Quality Checks	5
4. Mitigations Applied	5
5. Impact Assessment on Analysis	5
6. Sensitivity and Robustness Checks	5
7. Appendices.....	5

1. Data Overview and Coverage

1.1 All source data sets – Complete inventory

#	Folder	File(s)	Rows	Used in Scripts	Notes
1	1. customer_transaction	7 yearly Excel files	153,828	EDA scripts 01-11	Primary source. FX conversion applied
2	2. product_master	2_1.products_master_20260505.xlsx	167	01_load_and_merge.py 02_data_quality.py	Saved to parquet. Used for category classification in DQ audit only.
3	3. Discounts	3_1.discounts_export_20260505.csv	367	01_load_and_merge.py 05_channel_discount.py	Discount code taxonomy. Cannot link to individual orders (no discount code field in transactions).
4	5.Recharge_data	5_1.orders_combined	1,215	06_subscription_churn.py	Apr 2025–Apr 2026 only. Avg order S\$81 (SGD)
5	5.Recharge_data	5_2.order_items_checkout	1,094	04_product_analysis.py 06_subscription_churn.py`	First-checkout SKU ranking
6	5.Recharge_data	5_3.subscribers_reactivated	50	06_subscription_churn.py	Win-back cohort
7	5.Recharge_data	5_4.subscriptions_churned	526	06_subscription_churn.py	Cancellation reasons and churn cycle
8	5.Recharge_data	5_5.order_items_recurring	650	04_product_analysis.py 06_subscription_churn.py	Renewal SKU loyalty analysis

1.2 Canonical output tables

#	Table	Rows	Key fields
1	orders.parquet	27,350	order_id, customer_id, order_date, store, revenue (SGD), discount (SGD), channel, is_subscription
2	lines.parquet	50,963	order_id, customer_id, product_category, Line: Price/Total (SGD)
3	customers.parquet	13,780	customer_id, first/second order dates, total_orders, total_revenue (SGD), RFM fields
4	products.parquet	167	Handle, Variant SKU, prices by market, Status, Cost per item
5	discounts.parquet	367	Name, Value, Value Type, Times Used, Status
6	rc_orders.parquet	1,215	metric_date, shopify_order_id, order_type, order_total (SGD est.)
7	rc_checkout.parquet	1,094	product_title, variant_title, quantity, customer_id (Recharge ID)

ev e	rc_reactivated.parquet	50	customer_id (Recharge ID), first_activation, reactivated_date
9	rc_churned.parquet	526	customer_id, cancellation_reason, subscription tenure
10	rc_recurring.parquet	650	product_title, variant_title, purchase_type

1.3 Temporal Coverage

Period	Orders	Unique Customers	Notes
2019	3	3	Incomplete year — excluded from cohort analysis
2020	2,847	1,696	Full year, zero discounting
2021	6,259	3,815	Peak revenue year (S\$848K)
2022	3,710	2,299	First discounting introduced
2023	2,253	1,399	Revenue collapse year
2024	4,261	2,621	Recovery, heavy discounting
2025	6,407	4,107	Highest order volume, continued discounting
2026 (Partial)	1,610	1,230	Jan–Mar only; excluded from full-year analyses

Period of analysis: 2019-12-31 to 2026-03-30

1.4 Store and currency coverage

Store	Order	Revenue (Own Currency)	FX Rate	Revenue (SGD equivalent)
SG	16,041	SGD \$1,913,387	1.00	S\$1,913,387
MY	11,309	RM \$3,958,566	3.30	S\$1,199,56
HK	2	HK\$1,943	6.10	S\$319
All markets	27,350			S\$3,112,952

Notes: The three stores use different currencies. The raw data contains no FX rates.

Mitigation applied (May 2026): 5-year average exchange rates (2020–2026) applied at data load time:

- 1 SGD = 3.30 MYR

- 1 SGD = 6.10 HKD

These are 5-year average rates (not spot rates at each transaction date). This introduces a measurement error in absolute revenue figures of up to $\pm 10\%$. The direction and relative magnitude of all findings are unaffected — retention, repeat rates, and channel comparisons are count-based and FX-neutral.

Known limitations of fixed-rate FX approach:

1. SGD/MYR moved between ~ 3.0 and ~ 3.5 over 2020–2026. Early-year MY revenue (2020–2021) may be understated/overstated by up to 10%.
2. MY market was effectively dormant by 2025 (only S\$9,617 SGD in 2025 vs S\$441,022 in 2021), so the combined totals are dominated by SG data in recent years.
3. HK store (2 orders, S\$319 SGD equivalent) is immaterial.

2. Summary Statistics and Distributions

3. Data Issues and Discrepancies

3.1 Mixed Currencies

Orders in the dataset comes from 3 different regions (Singapore, Malaysia, and Hong Kong) which uses different currencies (SGD, MYR, HKD). The raw data contains no FX conversion rates at the point of sale which makes it hard to analyse the sales data across all the countries. This data issue has a high impact on the analysis. Without converting the currencies, it would cause the revenue totals to be inaccurate as the different exchange rates will not be considered and the value of all currencies will be assumed the same.

Since it is difficult to get the currency conversion rate on the day of the transaction, the average exchange rate for the past 5 years was used to convert the currencies all to SGD to solve this issue. For MYR, the rate of 1 SGD = 3.3 MYR was used and for HKD, the rate of 1 SGD = 6.1 HKD was used.

3.2

4. Mitigations Applied

5. Impact Assessment on Analysis

6. Sensitivity and Robustness Checks

7. Appendices